Jago Alcock

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Graduate mechanical engineer with 2+ years of research and development work experience across design, prototyping, testing, and manufacturing. Impassioned by finding creative solutions to design, automation, and optimisation problems. An avid sportsperson, having represented New Zealand as a Junior Paddle Black and spending my free time mountain biking, surfing, hiking and in the snow.

Work Experience

Junior Engineer – SYOS Aerospace (aerospace - 3 months)

SYOS Aerospace develops advanced uncrewed air, ground, sea and subsurface systems for defence and commercial applications. Unfortunately, due to the nature of this work, some project details are classified.

- Designed and manufactured components for customer-ready UAS products, working to tight tolerances and delivery standards.
- Sole troubleshooter on a customer contract following commissioning issues. I diagnosed and resolved faults across systems under time pressure.
- Designed a ~12×12m drone test enclosure to allow safe indoor UAS flight testing, protecting both aircraft and personnel in the event of a crash.
- Redesigned a customer-facing Ground Control Station (GCS) to improve structural integrity, transmission reliability, and operator usability.
- Supported testing, manufacturing, and general operations, contributing wherever needed in a fast-moving environment.
- Produced technical documentation in line with customer and project requirements.

Junior Engineer – SPS Automation (aerospace - 3 months)

- Developed a Wireless Aerial Nozzle Device (W.A.N.D) for drone-mounted precision spraying, from initial concept through to a flight-tested proof-of-concept product.
- Designed a dual-axis nozzle actuator with ±90° simultaneous vertical and horizontal actuation, integrating LiDAR and an IP camera for real-time range and visual feedback.
- Provided a flexible, robust and easy to use stand-alone spray system, engineered to SPS Automation's standard mounting interface, enabling compatibility across multiple aerial platforms.
- Delivered comprehensive documentation covering design rationale, technical specifications, assembly, SOPs, and recommended improvements.

Junior Engineer – Tait Communications (critical communication infrastructure - 3 months)

- Redesigned a PCB enclosure and mounting system for the software development team, replacing a dangerous and space-inefficient existing solution.
- Led full design process: customer research, concept development, iterative prototyping, manufacturing, and testing.
- Produced detailed handoff documentation to ensure continuity of the project after departure.
- Collaborated and learnt from diverse, multi disciplined teams.

Junior Engineer – Gyro Plastics (electrical infrastructure - 6 months)

- Conducted R&D on 10+ new and existing products guiding ideas through every development stage.
- Facilitated the design and implementation of workshop process and equipment improvements.
- Developed impact protection bollards all the way through to manufacturing processes and production units.
- Designed EV charging kiosks, balancing wind loading, tamper resistance, ease of installation, and mould fabrication requirements.
- Managed a hybrid remote/in-person work model, maintaining full team integration across two locations with Gyro Plastics arranged fortnightly return flights from Canterbury to Manawatu.

CNC Machining – Automatic Lathes (manufacturing - 3 months)

- Set up, operated, and maintained CNC mills and lathes in a production environment. Additionally responsible for cutting, deburring, quality checking, and workshop upkeep.

Research and Development – QuickBuild Homes (housing - 12 months)

- Contributed across machine programming, 3D modelling, product design, 3D printing, factory work, graphic design, conceptual site plans, 3D rendering and visualisation for prefab housing manufacture.

Responsibilities held across roles include:

- **Product research and feasibility** (costing, competitor, and standards research).
- **Concept development and design** (sketching, 3D modelling, and 3D rendering).
- **Rapid prototyping** (3D printing, CNC machining, laser cutting, folding).
- **Testing** (load testing and system functionality testing of physical prototypes).
- **Implementation into production** (rotational moulding, CNC machining, 3D printing and outsourced manufacture).
- **Project management** (managing lead times, setting deadlines, keeping time sheets)
- **Collaboration** (with manufacturing, software, electrical, mechanical teams).
- **Documentation** (to facilitate seamless handoffs of my projects).

Technical Skills

- **Engineering software:** SolidWorks, ANSYS, onshape, Fusion360, Creo, Mastercam, VISI, Archicad, Python.
- **CNC tools:** 3D printers, mills, lathes, saws.
- **Manual tools:** Lathes, bandsaws, drill presses, soldering irons, hand tools.
- **Visualisation software:** Photoshop, Lumion.
- **Driver's license:** Class 1
- **CAD, CAM, CNC Certificates:** https://cncexpert.com/jago-alcock

Personal Skills

- **Fast Learning:** Picked up and trusted with new responsibilities quickly across every role, despite limited prior experience. One of the qualities I am most proud of.
- **Problem Solving:** I have successfully tackled a wide range of design challenges using diverse techniques.
- **Communication and Teamwork:** Lifelong involvement in international team sport has sharpened my ability to communicate and collaborate effectively, including under pressure.
- **Leadership**: I have held various leadership roles (House Captain, Sports Captain, Captain of the NZ U21 Canoe Polo team).

Achievements

- Vice Chancellor's Excellence Scholarship
  For high-achieving candidates who have a record of academic excellence, leadership, and community engagement.
- Contel Charitable Trust Masters Scholarship
  For general character, leadership potential and communication skills, academic achievement.
- Caliber Design Prize in Mechanical Engineering
  For excellence by BE(Hons) students of Mechanical Engineering at University of Canterbury.
- CWF Hamilton Masters Scholarship
  For academic achievement and potential to contribute to research in Mechanical Engineering.
- Ken Whybrew Memorial Prize
  For excellence in Manufacturing Technology.
- Courtney Shearer Memorial Scholarship in Engineering
  For engineering students, with a preference for those who did not come to university directly from school and who worked in the intervening period.
- Academy of Sport Scholarship
  For new student-athletes with a previous record of high achievement in sport and who show future potential.
- Blues Award (x2)
  Celebrates students who have excelled in sport, arts, and community engagement.
- Principal's Excellence Award
  Awarded for commitment to academic success.

Community

2025: Tutor - Controls and Vibrations

- Taught undergraduate students in the Controls and Vibrations course at the University of Canterbury.

2024: Team Leader - UC Aerospace Level 1

- Led a team in building and launching a 1.4m rocket kit. I gained practical experience in rocket aerodynamics, fabrication, and launch procedures.

2016 – 2025: Canoe Polo

- Represented New Zealand as part of the NZ U21 Men's team: Gold at 2019 Junior International Championship (Northern Ireland); Bronze 2022; 6th at 2022 World Championship (France).
- Captained the NZ U21 Men's team to 1st place at the 2023 Oceania Championship.
- Volunteered as referee and coach at club level.

Referees

Jake Tisdale – SYOS Aerospace

- Mechanical Department Lead
- jake.tisdale@syos-aerospace.com
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Trudi Duncan – Gyro Plastics

- Managing Director
- trudi.duncan@gyroplastics.com
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Jordie Peters - MotorSport NZ / Liam Lawson Motorsport

- Digital/ Media Manager
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